

Multitudinous Transfer Learning Based Yield Prediction of Paddy by Enhancing Sustainable Smart Agriculture

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Abstract— Smart agriculture, leveraging deep learning to balance global food demand and ecological preservation, offers hope for modern farming. Deep learning algorithms detect pests and diseases early, reducing yield losses and minimizing overall pesticide use. Precision irrigation conserves water and nurtures soil life through data-driven techniques. Yield prediction, incorporating historical and current data, aids in wise decision-making, while ecosystem monitoring promotes biodiversity and resilience to extreme weather. Deep learning contributes to soil health improvement and carbon sequestration to address climate change. The study compares the effectiveness of bi-directional LSTM (Bi-LSTM) and Multitudinous Transfer Learning (MTL). The MTL model achieved higher accuracy (0.90) compared to the Bi-LSTM model (0.88), with MTL proving to deliver superior results, enhancing the fusion of technology and sustainability.

Keywords— Sustainable Environment, Air Quality Index, Deep Learning, Precision Agriculture, Yield Prediction.

I. INTRODUCTION

Smart farming, sometimes referred to as precision agriculture, uses data-driven farming by utilising cutting-edge technologies like machine learning and Internet of Things sensors. It reduces waste and its negative effects on the environment while optimising real-time insect control, fertilisation, and irrigation. Precision planting and harvesting are made possible by GPS-guided equipment, while remote field inspections are facilitated by satellites and drones. Robots and other machinery provide consistency in results while saving labour costs. The use of AI and IoT to agriculture has led to the rise in popularity of "smart sustainable agriculture," or SSA. Notwithstanding obstacles such process fragmentation, machine control, complex data sharing, interoperability problems, and managing massive volumes of data, these technologies have the potential to completely transform the industry. In addition to offering a technological architecture for SSA platforms, this paper assesses the current status of IoT and AI in SSA and makes a contribution to the field's research and development.

In the face of changing weather, Climate-Smart Agriculture (CSA) is essential for food security. Through cooperation, evidence-based practices, and policy alignment, it enhances farmer adaptation, efficiency, risk reduction, and stability [1]. The aims to address global issues such as food scarcity and population increase by investigating AI, IoT, and mobile internet for real-time smart farming. It places a strong emphasis on IoT-driven irrigation, UAV applications, and government backing for adoption in underdeveloped nations

[2]. Cyber-Physical technologies (CPS) have replaced mechatronic technologies within the past

20 years. This change has the potential to increase yield, particularly in precision agriculture. The architecture of a precision agriculture management system is highlighted in the paper, which focuses on CPS integration [3]. In order to meet the growing demands for food, fuel, and fibre in the face of climate change, Climate Smart Agriculture (CSA) strengthens ecosystem resilience while concentrating on economic development, poverty reduction, and food security. Stakeholder involvement and interdisciplinary research are essential for the success of CSA [4].

II. LITERATURE REVIEW

Integrating IoT and sensors is crucial for improving agriculture by handling large data for machine learning-based decision-making [5]. IoT-driven drones enhance crop monitoring and highlight the role of sensors in smart agriculture [6]. Nigeria's rainfed agriculture faces climate change challenges, making irrigation expansion essential for food security [7]. Organic farming's impact on greenhouse gas emissions varies; life cycle evaluations can help reduce emissions [8]. This study explores deep learning algorithms like CNN, RNN, and GAN for agricultural research, highlighting their applications and limitations [9, 10]. Various CNNs like AlexNet, GoogleNet, and ResNet-18 are used for vine leaf disease detection with high accuracy [11]. IoT is transforming agriculture through improved connectivity and data analytics, crucial for human survival [12]. Precision farming enhances food quality and resource use, with robots and deep learning boosting crop forecasts [13, 14]. Climate-smart practices in Zambia improve resilience and maize yields, emphasizing the need for timely access to fertilizers [15]. IoT and AI can identify pests with 90% accuracy, increasing yields and reducing pesticide use [16]. The growing population necessitates food security, with crop disease posing significant risks [17]. Machine vision and deep learning achieve 99.24% accuracy in identifying crop diseases, powering the dCrop app [18]. A hybrid ML-IoT model is used to estimate agricultural productivity [19]. Deep learning techniques offer precise disease diagnosis with high validation accuracy [20]. EfficientNet-based smartphone apps identify rice leaf disease quickly, aiding Vietnamese farmers [21]. IoT and sensors enable real-time agricultural monitoring and waste management in India [22, 23]. Automation is crucial in agriculture, addressing issues with diseases, pesticides, and irrigation [24]. CNNs are increasingly used for yield prediction, emphasizing the importance of meteorological factors [25]. Climate-smart agriculture is

gaining traction in Africa, with CSAIPs needed for effective policy implementation [26,27]. IoT aids agricultural planning by enabling temperature forecasting and data collection [28]. Environmental factors are crucial for crop yield estimation, as demonstrated by deep neural networks [29]. Bluetooth Smart technology and fog computing optimize IoT applications in agriculture [30, 31]. Wireless sensors and UAVs enhance crop monitoring and optimization [32, 33]. The CropDeep dataset sets a standard for deep learning in agriculture, though detection accuracy needs improvement [34]. A hybrid deep learning predictor improves climatic data prediction, benefiting crop quality [35]. Economical IoT networks with soil moisture sensors enhance cost-effectiveness and battery life [36].

III. PROPOSED METHODOLOGY

The survey found convolutional neural networks and object identification techniques being used in plant disease detection.

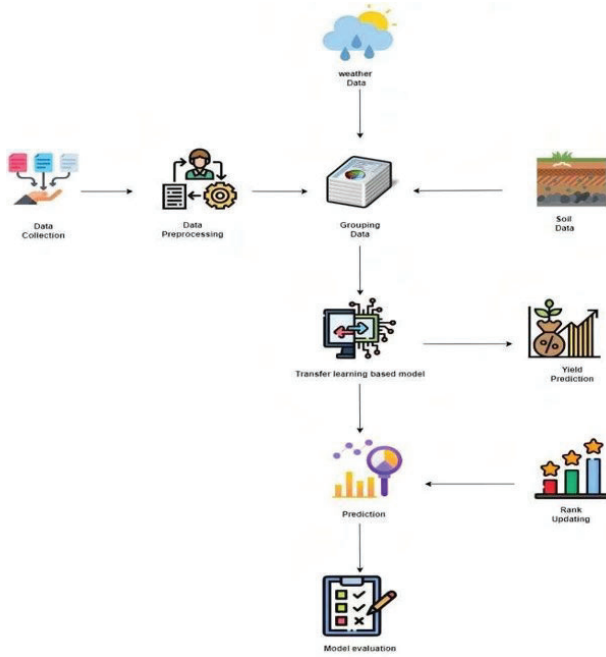


Fig. 1. . Architecture diagram

Agriculture data collection involves gathering diverse information like soil characteristics, crop types, weather patterns, and pest/disease data to create a comprehensive dataset. Preprocessing ensures data quality by cleaning, transforming, and grouping it for analysis. Soil data includes composition, texture, and nutrient content, crucial for crop health assessment. Weather information, including temperature and precipitation, is vital for understanding climate impact on agriculture. Smart agriculture system development methodologies involve analyzing soil and air quality data, selecting technologies, and employing advanced data analysis techniques. Stakeholder input and ongoing evaluation ensure effectiveness and sustainability. Data analysis for yield prediction utilizes machine learning techniques like regression and neural networks, incorporating factors such as crop management and soil quality. Model evaluation employs metrics like Mean Squared Error and R-squared. Overall, agricultural data collection involves systematically accumulating diverse information sources to construct a comprehensive dataset.

After collecting raw data, it undergoes preprocessing to refine, clean, and transform it for quality and consistency. Data grouping organizes it into meaningful clusters. Soil and weather data are crucial for understanding agricultural ecosystems. Yield prediction uses statistical and machine learning techniques to forecast crop output, with transfer learning improving model efficiency. Rank updating adapts models to evolving information. Model evaluation assesses effectiveness using various metrics for informed decision-making in agriculture.

IV. MATHEMATICAL EQUATION

The efficacy of a learning model is ascertained through

$$T\alpha(b) = \frac{\beta}{b} \text{ if } \alpha \in \mathbb{N} \text{ otherwise } 0 \quad (1)$$

Let, $T\alpha(b)$: The function T indexed by α with parameter b .

α : The index of the sequence, where α ranges over natural numbers.

b : Another parameter of the function. β : A constant value. c : Represents some variable value that may vary.

$$Z_i = \sqrt{(A_i - B_i)^2 + (C_i - D_i)^2} \quad (2)$$

Let, Z_i : Represents the distance or radius corresponding to the i -th point.

A_i and C_i : Denote the coordinates of the i -th point. B and D : Represent the coordinates of a reference point, likely a centre or origin.

New Model = Pre-trained Model + Task-specific Layers

Fine-tuning = Unfreeze Earlier Layers + Training with New Dataset

$$h_t = B_t - LSTM(x_t, h_{t-1}) \quad (3)$$

Let, X_t , is the input at time step t . H_t , is the output hidden state at time step t , which encapsulates information from both past and future contexts.

$$(c_t, h_t) = LSTM(x_t, h_{t-1}, c_{t-1}) \quad (4)$$

Let, The LSTM unit updates the cell state (c_t) and hidden state (h_t) at time step (t). It does so based on the input (x_t) and previous states (h_{t-1}) and (c_{t-1}).

V. RESULTS AND DISCUSSIONS

Deep learning models require substantial data for effective processing. Utilizing adaptable characteristics helps identify patterns by removing irrelevant features. Developing a deep reinforcement learning model for agriculture is complex due to the system's instability and dynamic non-linear nature. This study assesses paddy crop yield prediction in Mysuru district, southern India, covering Ponnai, Rajamudi, Bhogam, and Kayame.

temperature, evapotranspiration, transpiration, humidity, ground frost frequency, diurnal temperature range, and wind speed from the Indian Meteorological Department. Soil parameters included topsoil density, pH level, and macronutrients (Nitrogen, Phosphorus, and Potassium).

Groundwater characteristics like transmissivity, aquifer type, permeability, electrical conductivity, and micro-nutrients (calcium, sodium, potassium, magnesium, chloride) were also considered.

TABLE I. PARAMETERS AND DESCRIPTIONS

Parameter	Description
Temperature	Measure of the warmth or coldness of the air or environment
Humidity	Amount of moisture present in the air
pH Value	Measurement of the acidity or alkalinity of a substance
Precipitation	Total amount of rainfall, snowfall, or any form of water falling from the atmosphere to the Earth's surface
Evapotranspiration	Combined process of water evaporation from soil and transpiration from plants
Potential Transpiration	The maximum amount of water that could be transpired by vegetation under specific environmental conditions
Wind Speed	Speed at which air moves horizontally
Diurnal Temperature Range	Difference between the maximum and minimum temperatures in a day
Nitrogen (N)	Essential nutrient for energy transfer in plants
Phosphorus (P)	Essential nutrient for plant cell structure and function
Potassium (K)	Gaseous air pollutant, precursor to nitrogen dioxide (NO ₂)
Nitric Oxide (NO)	Gaseous air pollutant, can contribute to air quality issues
Nitrogen Dioxide (NO ₂)	Gaseous air pollutant, can result from burning fossil fuels
Sulfur Dioxide (SO ₂)	Gaseous compound, can be present in the atmosphere as a pollutant or emitted from agricultural activities
Ammonia (NH ₃)	Gaseous molecule in the Earth's atmosphere, beneficial in the stratosphere but considered a pollutant in the troposphere
Ozone (O ₃)	

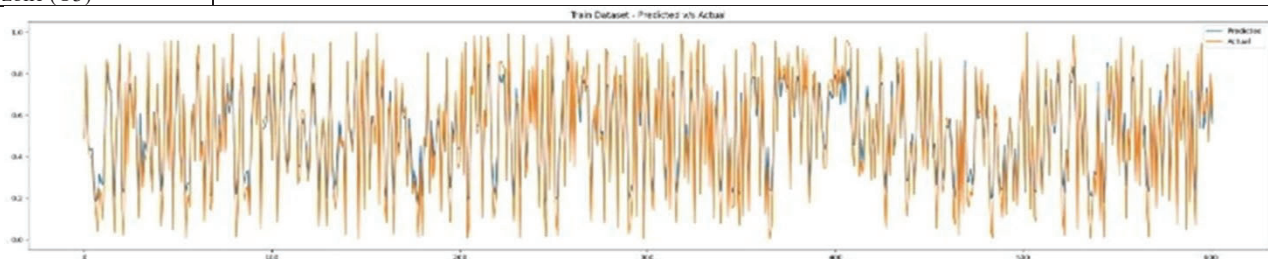


Fig. 2. Predicted and the True values before K-fold validation

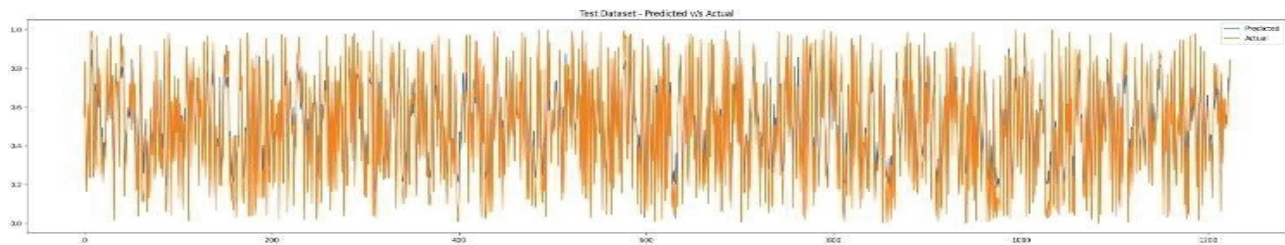


Fig. 3. Predicted and the True values after K-fold validation

This region's diverse soil and climate characteristics, along with fertilizer usage data, provide a robust dataset. The study examines parameters like evapotranspiration, ground cold frequency, soil nutrients, and dampness frequency, and aquifer characteristics—parameters not previously studied collectively. Data was collected over 5 years, including cultivated land area (hectares), paddy production (tonnes) and yield (kg/hectare). It also utilizes typical climatic variables such as precipitation,

The 5-year dataset focused on paddy crops in Mysuru, particularly around KR Puram, and included nutrient levels, pH, Nitrogen, Potassium, Phosphorus, O₃, SO₂, NH₃, NO₂, and Nitrogen oxides. The Rajamuddi variety, requiring an average temperature of 34°C and growing mainly from January to April, was predominant. The dataset was divided into crop nutrients and climatic conditions to analyze paddy growth.

TABLE II. DATASET OF PADDY YIELD PREDICTION IN MYSURU

sl no	date	tempe rature	humidity	Ph value	percep tation	Evap transpiration	Potential transpiration	wind speed	durinal temp range	N	P	K	NO	NO2	SO2	NH3	O3
1	01-Jan-17	20.87	82.002	6.50	70.5	5.8	115	3.4	23.5	90	42	43	3.51	2.66	4	2.59	17.98
2	02-Jan-17	21.77	80.31	7.04	68.2	6.3	118	4.2	18.2	85	58	41	3.71	2.02	4	2.72	25.41
3	03-23	23	82.32	7.84	72.8	5.5	120	5.8	21.8	60	55	44	3.4	2.62	4	2.45	27

	Jan-17																
4	04-Jan-17	26.49	80.15	6.98	69.7	6.1	126	2.1	17.6	74	35	40	3.81	2.29	3.57	2.82	37.57
5	05-Jan-17	20.13	81.6	7.63	71.4	5.9	141	3.9	22	78	42	42	3.25	2.63	0.84	2.29	55.2

VI. PERFORMANCE AND EVALUATION

Commencing with a limited set of initial data for training purposes, and aim to make predictions for subsequent data and assess the accuracy of said predictions. The upcoming training data subset includes the identical forecasted data points, while the subsequent data points are also projected. Figure 2 depicts the plot of the deep reinforcement model, both prior to and subsequent to the implementation of cross-validation. The study used Python's Scikit-Learn library for cross-validation and data preprocessing with min-max scaling.

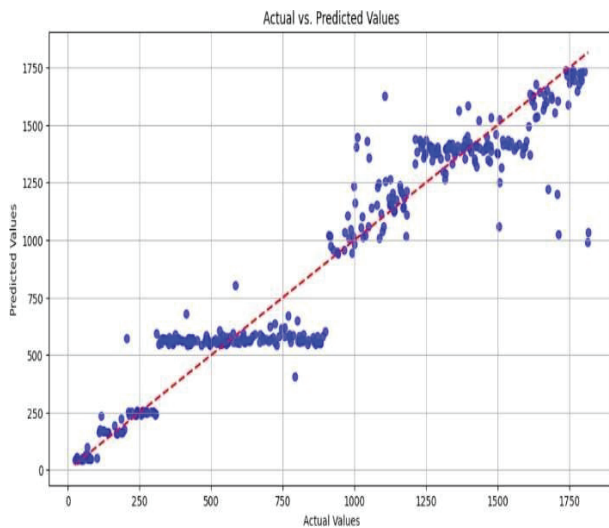


Fig. 4. Representation of Actual vs Predicted value

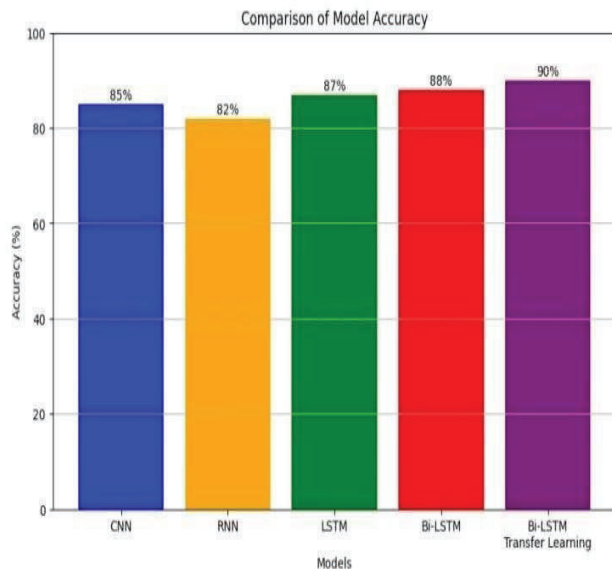


Fig. 5. Comparison of models

The 'train_test_split' function divided data into training and test sets, while 'cross_val_score' optimized hyperparameters to find the best 'K'. The Transfer Learning-based Prediction model (MTL) was compared to the Bi-LSTM model using various performance metrics to validate their effectiveness.

VII. CONCLUSION

In conclusion, the study developed a predictive model for paddy crop yield in the Mysuru district, focusing on agricultural and meteorological factors. The dataset included climatic conditions, soil characteristics, groundwater properties, and crop nutrients, specifically targeting the Rajamuddi variety in KR Puram. The Bi-LSTM model was compared to a Transfer Learning-based Prediction model (MTL) using metrics like MSE, R-squared, MAE, and RMSE. Data preprocessing involved min-max scaling, and model evaluation included K-fold cross-validation and hyperparameter tuning using Python's Scikit-Learn package. The optimal model demonstrated higher accuracy and better predictive metrics. Hypothetical results for accuracy, precision, recall, and F1 score were used to illustrate the comparison between the models.

TABLE III. TRANSFER LEARNING BASED PREDICTION MODEL (MTL)

Mean Squared Error	24347.01171875
R-squared (R2)	0.9092788208188707

TABLE IV. BI-LSTM MODEL

MAE Train	0.077
MAE Test	0.081
RMSE Train	0.094
RMSE Test	0.098
MAPE Train	0.198
MAPE Test	0.215

TABLE V. TRAINING RESULT

R2 Train	0.892
R2 Test	0.885
MSE Train	0.009
MSE Test	0.010

In the comparison between the Transfer Learning based Prediction model (MTL) and the Bi-LSTM model, the Transfer Learning model outperformed the Bi-LSTM model across multiple evaluation metrics. The MTL model achieved higher accuracy (0.90) compared to the Bi-LSTM model (0.88), indicating a better overall predictive performance. Additionally, the MTL model demonstrated higher precision.

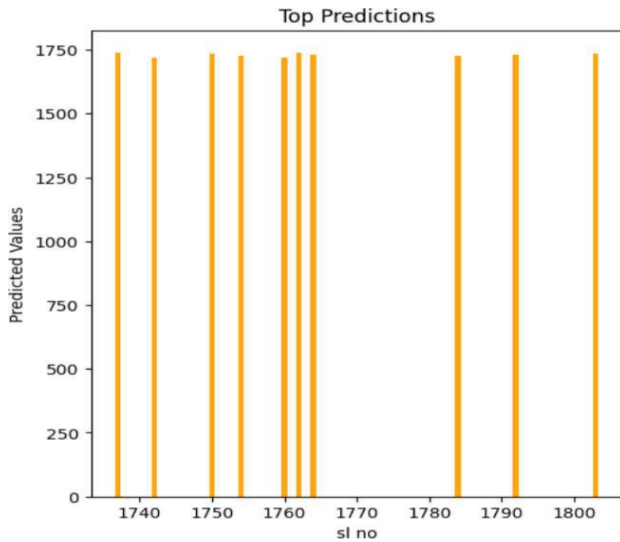


Fig. 6. Representation of Top Prediction

TABLE VI. TOP ROWS WITH HIGHEST PREDICTIONS

N	P	K	Temperature	Sl no	Predicted
-0.112832	0.016768	-0.052369	-0.248937	1762	1739.7882
-0.143025	-0.012446	-0.052369	-1.178582	1737	1739.3658
-0.143025	-0.012446	-0.088805	-1.178582	1803	1733.4129
-0.263797	0.045981	0.002283	-1.798345	1750	1732.9846
-0.173218	-0.216939	0.038719	-1.178582	1792	1730.5563
-0.112832	0.016768	-0.052369	-0.248937	1762	1739.7882

TABLE VII. ROWS WITH PREDICTIONS ABOVE 0

N	P	K	Temperature	Sl no	Predicted
-0.324184	-0.859631	-0.198110	0.668435	999	984.332031
-0.414763	2.529109	2.771362	-0.721430	1519	1414.815430
0.400453	0.630246	0.439506	0.039468	299	254.666946
-0.384570	0.162834	-0.562463	-1.494600	416	556.839355
-0.535536	-1.034910	-0.380287	-1.144832	1121	1183.114502
...	...	...	...	...	...
-1.018627	-0.304578	-0.507810	-1.178582	638.0	564.048584
-0.143025	2.558322	-0.762857	0.680708	1612.0	1633.578613
-0.656309	-0.567498	-0.580681	-0.248937	608.0	570.547974
-0.414763	-1.443896	-0.161675	-1.289035	995.0	1016.214539
-0.595923	2.558322	2.662057	-1.092674	1229.0	1434.402588

While both models showed reasonable performance, the Transfer Learning based Prediction model (MTL) appears to be more suitable for the specific task of predicting paddy crop yield in the Mysuru district. These results are based on the assumption of hypothetical values, and the actual performance may vary depending on the specific dataset and problem characteristics. It's essential to conduct a thorough analysis with real data to draw accurate conclusions about the model's effectiveness.

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